SQL Interview Questions and Answers

1. What is SQL?

Interviewer: What is SQL?

Candidate: SQL stands for Structured Query Language. It is a standard language used to communicate with relational databases. Using SQL, we can create databases, insert data, update records, delete records, and retrieve information efficiently. In data analytics, SQL is one of the most important skills because it helps us extract meaningful insights from large datasets.

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2. What is a Database?

Interviewer: What is a database?

Candidate: A database is an organized collection of related data that is stored electronically. It allows users to store, retrieve, update, and manage information efficiently. For example, an e-commerce website stores customer details, product information, and orders in a database.

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3. What is DBMS?

Interviewer: What is DBMS?

Candidate: DBMS stands for Database Management System. It is software used to create, manage, and maintain databases. It provides features like data storage, security, backup, and recovery. Examples include MySQL, Oracle, SQL Server, and PostgreSQL.

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4. What is RDBMS?

Interviewer: What is RDBMS?

Candidate: RDBMS stands for Relational Database Management System. It stores data in the form of tables consisting of rows and columns. Tables are connected using relationships, making data retrieval efficient. MySQL and PostgreSQL are examples of RDBMS.

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5. What is the difference between DBMS and RDBMS?

Interviewer: Can you differentiate between DBMS and RDBMS?

Candidate: Yes.

- DBMS stores data but may not support relationships between tables.
- RDBMS stores data in related tables.
- RDBMS supports primary keys, foreign keys, and normalization.
- RDBMS is generally more secure and efficient for large applications.

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6. What is a Table?

Interviewer: What is a table?

Candidate: A table is a collection of related data arranged in rows and columns. Each row represents a record, and each column represents an attribute. For example, an Employee table may contain Employee_ID, Name, Department, and Salary.

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7. What is a Primary Key?

Interviewer: What is a Primary Key?

Candidate: A Primary Key is a column or combination of columns that uniquely identifies each record in a table. It cannot contain NULL values or duplicate values. For example, Employee_ID can be the primary key in an Employee table.

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8. What is a Foreign Key?

Interviewer: What is a Foreign Key?

Candidate: A Foreign Key is a column that creates a relationship between two tables. It references the Primary Key of another table and helps maintain data integrity.

For example, Department_ID in the Employee table can reference Department_ID in the Department table.

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9. What are Constraints?

Interviewer: What are SQL Constraints?

Candidate: Constraints are rules applied to table columns to ensure data accuracy and integrity.

Common constraints include:

- PRIMARY KEY
- FOREIGN KEY
- UNIQUE
- NOT NULL
- CHECK
- DEFAULT

These constraints prevent invalid data from being inserted.

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10. What is the SELECT Statement?

Interviewer: What is the SELECT statement?

Candidate: The SELECT statement is used to retrieve data from one or more tables.

For example:

SELECT * FROM Employee;

It is the most commonly used SQL command for querying data.

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11. What is the WHERE Clause?

Interviewer: What is the WHERE clause?

Candidate: The WHERE clause filters records based on a condition.

For example:

SELECT * FROM Employee

WHERE Salary > 50000;

This query retrieves only employees whose salary is greater than 50,000.

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12. What is ORDER BY?

Interviewer: What is ORDER BY?

Candidate: ORDER BY is used to sort query results in ascending (ASC) or descending (DESC) order.

For example:

SELECT * FROM Employee

ORDER BY Salary DESC;

This displays employees from the highest salary to the lowest.

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13. What is GROUP BY?

Interviewer: Explain GROUP BY.

Candidate: GROUP BY groups rows having the same values so aggregate functions can be applied.

For example:

SELECT Department, COUNT(*)

FROM Employee

GROUP BY Department;

This shows the number of employees in each department.

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14. What is HAVING?

Interviewer: What is the HAVING clause?

Candidate: HAVING filters grouped data after the GROUP BY operation.

For example:

SELECT Department, COUNT(*)

FROM Employee

GROUP BY Department

HAVING COUNT(*) > 5;

It returns only departments with more than five employees.

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15. What are Aggregate Functions?

Interviewer: What are Aggregate Functions?

Candidate: Aggregate functions perform calculations on multiple rows and return a single value.

Common aggregate functions include:

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

These functions are widely used in reporting and analytics.

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16. What is the Difference Between WHERE and HAVING?

Interviewer: Differentiate between WHERE and HAVING.

Candidate: WHERE filters rows before grouping, while HAVING filters groups after grouping.

For example:

- WHERE Salary > 30000
- HAVING COUNT(*) > 10

This is a common SQL interview question.

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17. What is a JOIN?

Interviewer: What is a JOIN?

Candidate: A JOIN combines data from two or more related tables using a common column.

For example, we can join Employee and Department tables using Department_ID to display employee names along with their department names.

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18. What are the Types of JOINs?

Interviewer: What are the different types of JOINs?

Candidate: The main types are:

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- FULL OUTER JOIN
- CROSS JOIN
- SELF JOIN

Each join retrieves data differently depending on business requirements.

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19. What is INNER JOIN?

Interviewer: Explain INNER JOIN.

Candidate: INNER JOIN returns only the matching records present in both tables.

For example, it displays employees who belong to an existing department.

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20. What is LEFT JOIN?

Interviewer: Explain LEFT JOIN.

Candidate: LEFT JOIN returns all records from the left table and matching records from the right table. If there is no match, NULL values are returned for the right table.

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21. What is the Difference Between DELETE, TRUNCATE, and DROP?

Interviewer: Differentiate DELETE, TRUNCATE, and DROP.

Candidate: Yes.

- DELETE removes selected rows and can use a WHERE clause.
- TRUNCATE removes all rows but keeps the table structure.
- DROP removes both the table and its structure permanently.

Choosing the correct command depends on the requirement.

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22. What is the Difference Between CHAR and VARCHAR?

Interviewer: Explain CHAR and VARCHAR.

Candidate: CHAR stores fixed-length strings, while VARCHAR stores variable-length strings.

For example:

- CHAR(10) always reserves 10 characters.
- VARCHAR(10) stores only the required number of characters.

VARCHAR is generally preferred because it saves storage.

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23. What is a View?

Interviewer: What is a View?

Candidate: A View is a virtual table created using a SELECT query. It does not store data itself but displays data from one or more tables.

Views improve security and simplify complex queries.

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24. What is an Index?

Interviewer: What is an Index?

Candidate: An Index improves the speed of data retrieval by allowing the database to locate records quickly.

Although indexes improve SELECT performance, they can slightly slow down INSERT, UPDATE, and DELETE operations because the index must also be updated.

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25. What is a Subquery?

Interviewer: What is a Subquery?

Candidate: A Subquery is a query written inside another SQL query.

For example, we can retrieve employees whose salary is greater than the average salary using a subquery.

Subqueries help solve complex problems efficiently.

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26. What is Normalization?

Interviewer: What is Normalization?

Candidate: Normalization is the process of organizing data to reduce redundancy and improve data integrity.

It divides data into related tables and establishes relationships between them.

The common normal forms are 1NF, 2NF, and 3NF.

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27. What is the Difference Between UNION and UNION ALL?

Interviewer: Differentiate between UNION and UNION ALL.

Candidate: UNION combines results from multiple queries and removes duplicate records.

UNION ALL combines results but keeps duplicate records.

UNION ALL is faster because it does not perform duplicate checking.

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28. What are SQL Clauses?

Interviewer: Name some commonly used SQL clauses.

Candidate: Some commonly used SQL clauses are:

- SELECT
- FROM
- WHERE
- GROUP BY
- HAVING
- ORDER BY
- LIMIT

These clauses help retrieve and manipulate data effectively.

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29. What are TCL, DCL, and DDL Commands?

Interviewer: Can you explain SQL command categories?

Candidate: Yes.

- DDL (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE
- DML (Data Manipulation Language): INSERT, UPDATE, DELETE
- DQL (Data Query Language): SELECT
- DCL (Data Control Language): GRANT, REVOKE
- TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT

Understanding these categories is important for SQL development.

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30. Why Should We Hire You for a SQL Role?

Interviewer: Why should we hire you as a SQL Developer or Data Analyst?

Candidate: As a fresher, I have a good understanding of SQL fundamentals, including database design, joins, constraints, aggregate functions, subqueries, views, and normalization. I have practiced writing queries on real-world datasets and understand how SQL is used in data analysis. I am eager to learn, enjoy solving data-related problems, and I am confident that I can quickly contribute to your team while continuously improving my SQL skills.